

2016.0 RANGE ROVER (LG), 501-29

SIDE PANEL SHEET METAL REPAIRS

'B' PILLAR OUTER PANEL

[G1532041]

REMOVAL AND INSTALLATION

REMOVAL

NOTES:

- The B-pillar outer panel is installed in conjunction with:
 - Rear door
 - Front seat
 - Front safety belt retractor
 - Headliner
 - Side air curtain module
 - Rear rocker panel moulding
 - Front scuff plate trim panel
 - Roof moulding
 - Right hand indicated, left hand similar.
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1.



E143329

The B-pillar outer panel is serviced as indicated.

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2. Before commencing this procedure make sure that you are aware of all Health and Safety requirements.

For additional information, refer to: Health and Safety Precautions (100-00 General Information, Description and Operation).

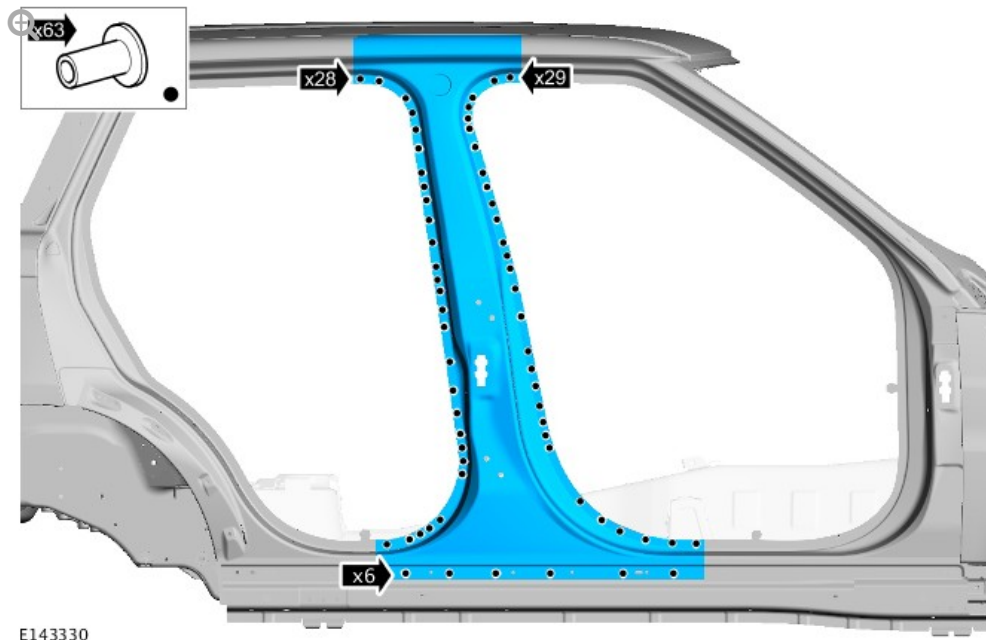
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3. For further information on the methods, tools and fixings used in this procedure refer to the body repairs - general information section.

NOTES:

- Where breakstem fasteners (BSF) removal is instructed remove fastener shaft with a 4mm punch and remove the head with a 6.5mm drill bit. Make sure all the drilled holes are free from debris.
- Where self piercing rivets (SPR) removal is instructed use either the approved SPR tool or 5.3mm drill bit.
- Any joints sealed as part of vehicle manufacture must be sealed as part of the repair.

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4. For additional information, refer to: Rear Door (501-03 Body Closures, Removal and Installation).
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5. For additional information, refer to: Front Seat (501-10 Seating, Removal and Installation).
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6. For additional information, refer to: Front Safety Belt Retractor (501-20 Safety Belt System, Removal and Installation).
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7. For additional information, refer to: Headliner (501-05, Removal and Installation).
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8. For additional information, refer to: Side Air Curtain Module (501-20 Supplemental Restraint System, Removal and Installation).
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9. For additional information, refer to: Rear Rocker Panel Moulding (501-08 Exterior Trim and Ornamentation, Removal and Installation).
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10. For additional information, refer to: Front Scuff Plate Trim Panel (501-05 Interior Trim and Ornamentation, Removal and Installation).
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11. For additional information, refer to: Roof Moulding (501-08, Removal and Installation).
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12. Remove the rear door upper and lower door hinges.
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13. Release the floor carpet and position it to one side.
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14. Release the B-pillar outer panel wiring harness and position it to one side.
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15.



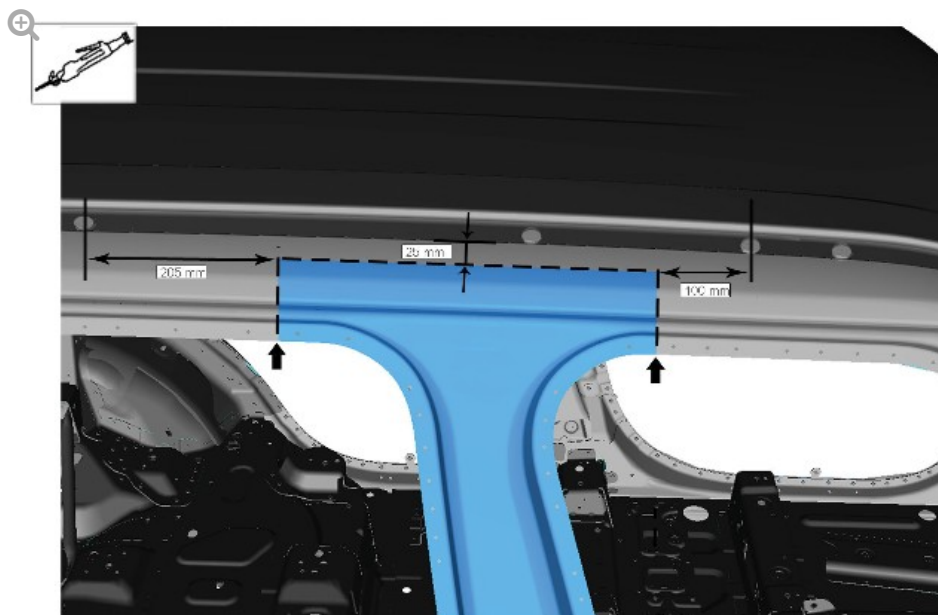
E143330

Remove the SPR's as indicated.

16.

CAUTION:

Care should be taken to avoid cutting through the inner panels and reinforcements.



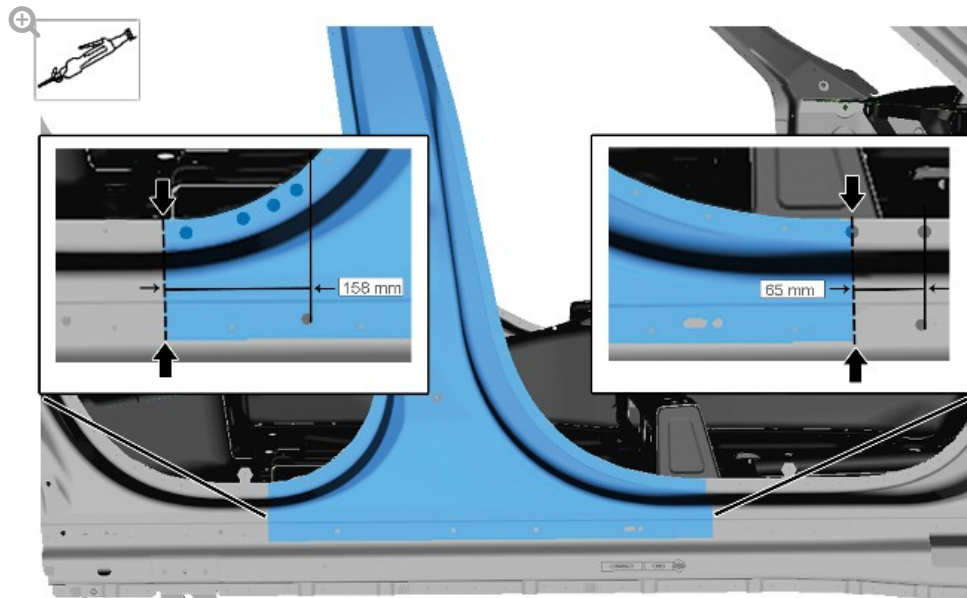
E143331

Mark out the positions where the metal inert gas (MIG) butt joints are to be made and cut through the B-pillar outer as indicated.

17.

CAUTION:

Care should be taken to avoid cutting through the inner panels and reinforcements.



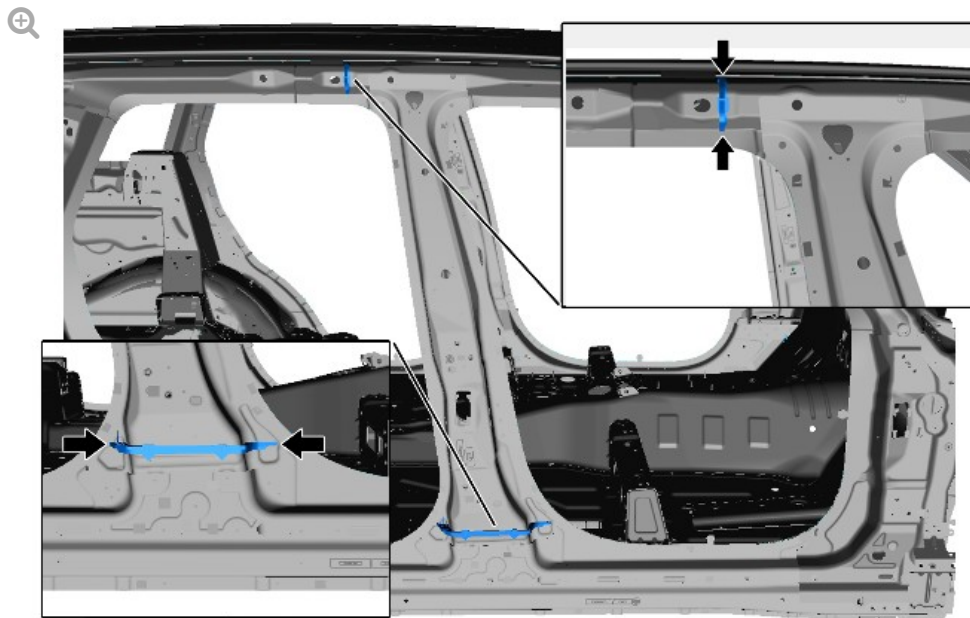
E 143332

Mark out the positions where the MIG butt joints are to be made and cut through the B-pillar outer as indicated.

18.

CAUTION:

Avoid damaging the noise, vibration and harshness (NVH) components that are located behind the B-pillar.



E143415

Separate the joints, remove the B-pillar outer panel and the NVH components.

INSTALLATION

CAUTION:

Where structural adhesive is to be applied between surfaces it is essential compression is applied to the joints during the curing period.

NOTES:

- BSF installation involves using the Land Rover approved BSF installation tool and requires a hole diameter of 6.5mm.
- SPR installation requires using the Land Rover approved SPR tool.

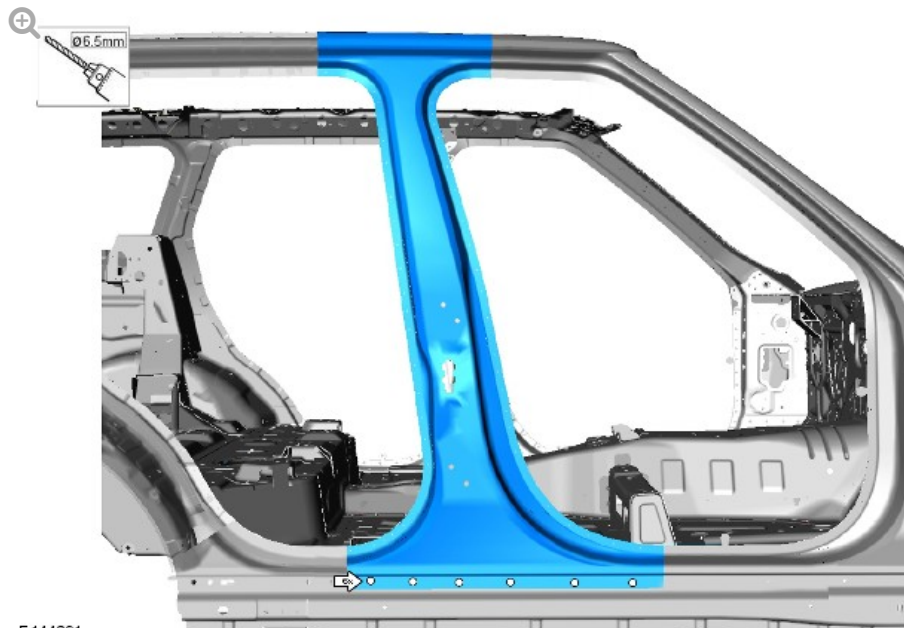
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1. Remove rivet remnants.
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2. Dress flanges where necessary.

3. Using a fine bristle disc, clean and prepare the panel surfaces.

4. Offer up the new B-pillar outer panel and clamp into position.
Check alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.

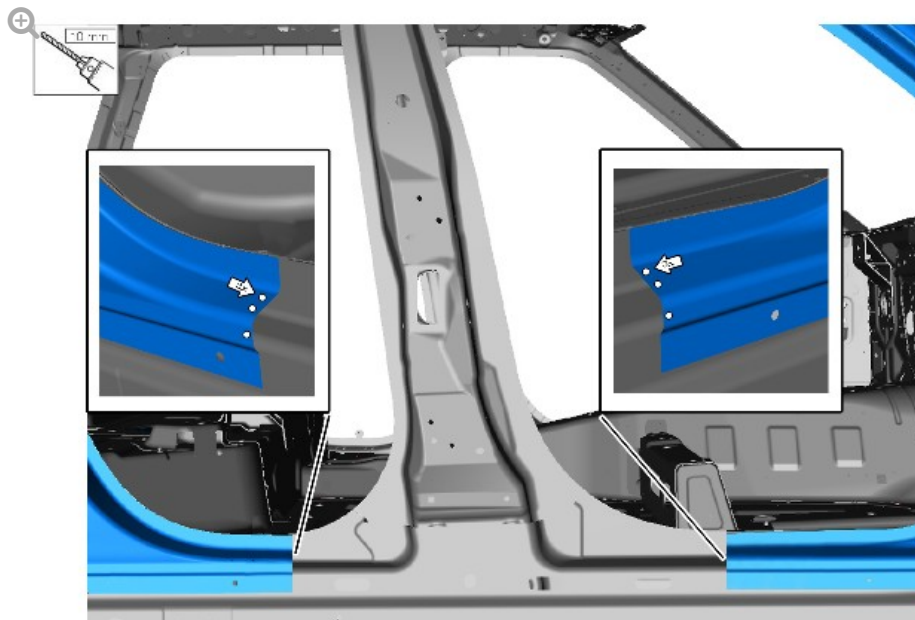
5.



Drill holes where BSF's are to be installed as indicated.

6. Remove the new B-pillar outer panel.

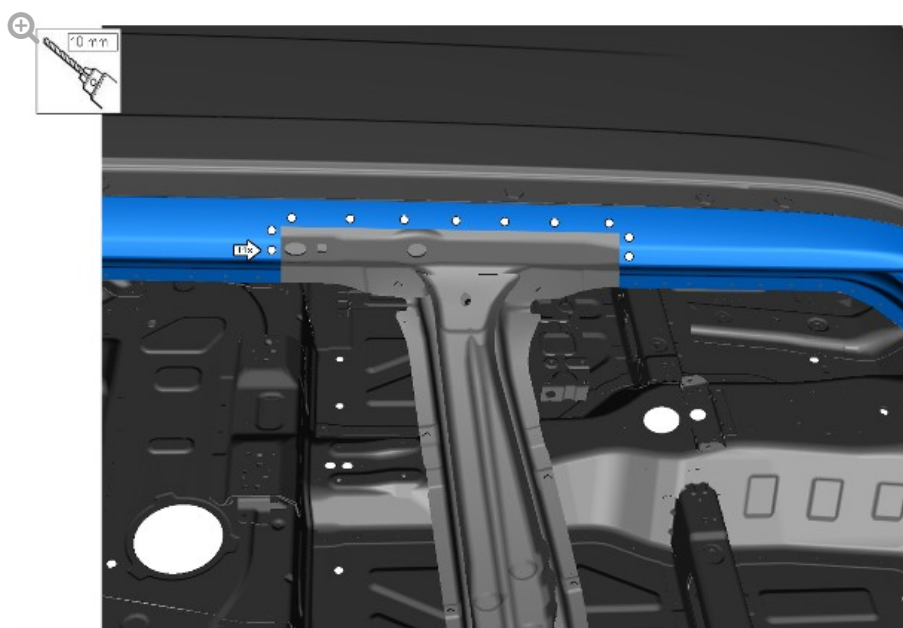
7.



E 143333

Using a 10mm drill bit, drill holes for the MIG plug welds ready for the installation of the backing plates as indicated.

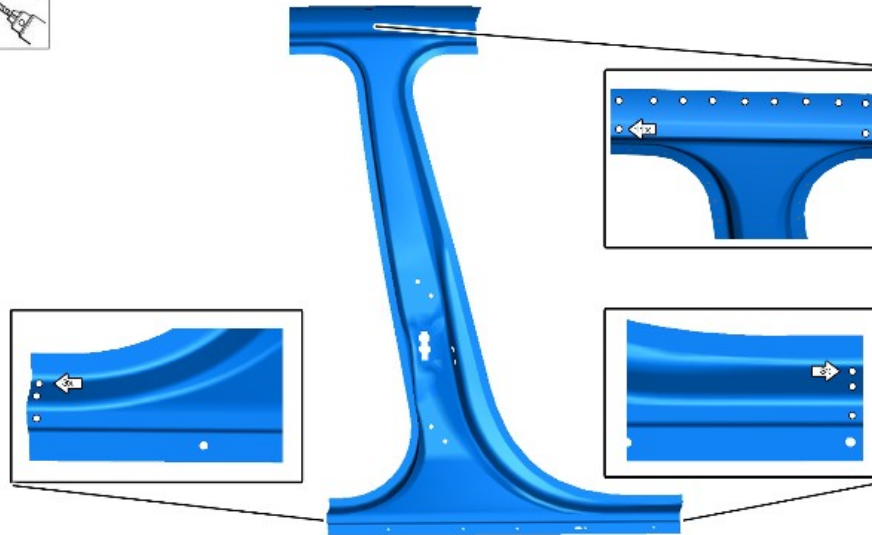
8.



E 143337

Using a 10mm drill bit, drill holes for the MIG plug welds ready for the installation of the backing plate as indicated.

9.

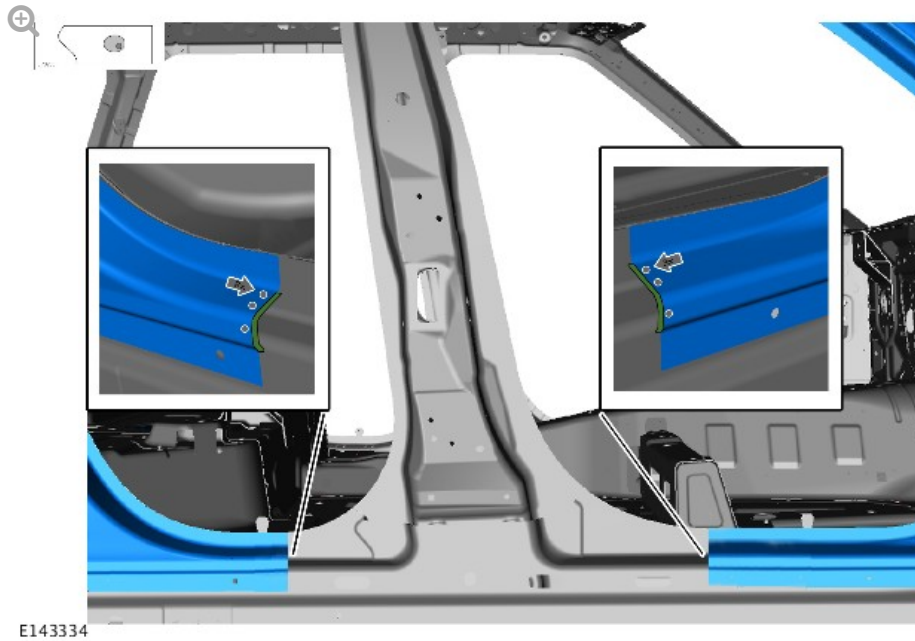


E143335

Using a 10mm drill bit, drill holes in the new B-pillar outer for the MIG plug welds ready for the installation of the backing plate as indicated.

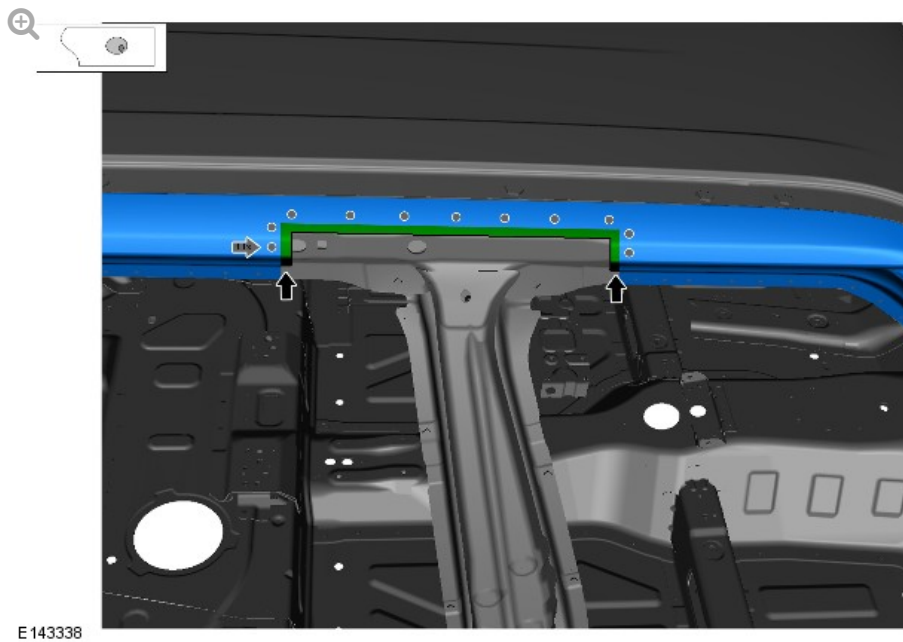
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10. Deburr the drilled holes.
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11. Measure and cut three 50mm wide backing plates, (25mm each side of the MIG butt joint), from the B-pillar outer panel remnant.
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12. Cut six run on/run off tabs from the B-pillar outer panel remnant.
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13. Using a fine bristle disc, clean and prepare the panel surfaces.
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14.



Install and MIG plug weld the backing plates as indicated.

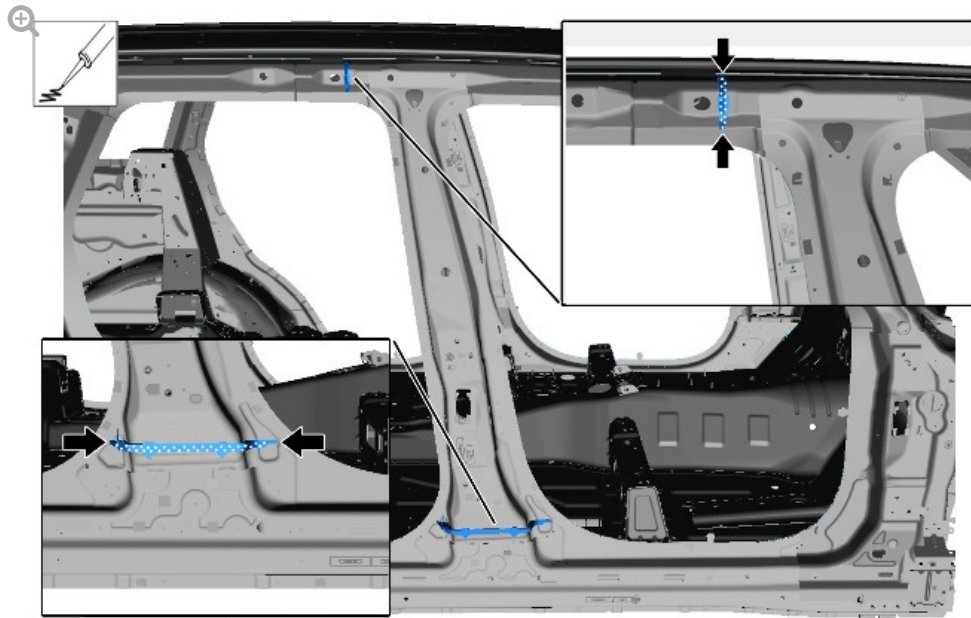
15.



Install and MIG plug weld the backing plate as indicated.

16. Apply the coupling agent where the Land Rover recommended bonding material is to be applied and allow to dry.

17.



E144292

Install the NVH components and apply a bead of Land Rover recommended semi-rigid sealer to the NVH components.

18.

NOTE:

Make sure a continuous bead of adhesive surrounds fixing holes.

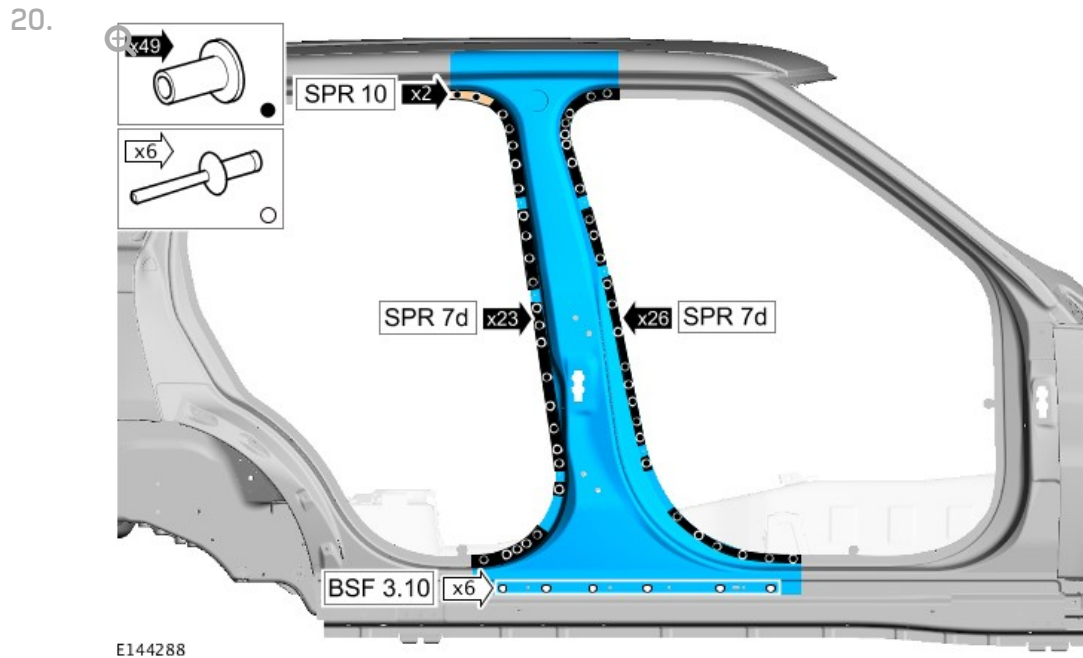


E143336

Apply a 5mm zig zag bead of Land Rover recommended bonding

material as indicated.

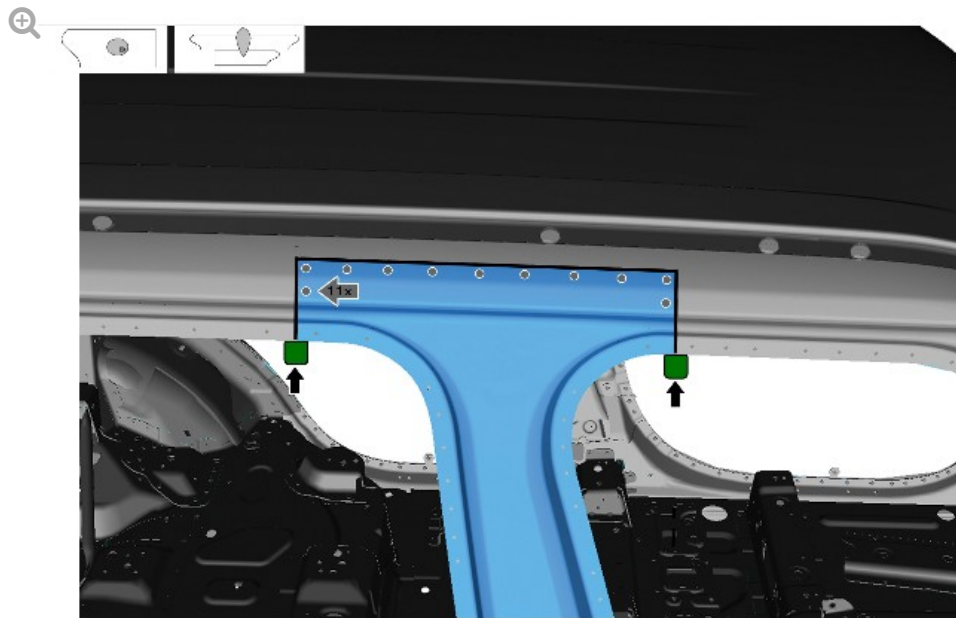
19. Offer up the new B-pillar outer panel and clamp into position.
Check alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.



Install the SPR's and the BSF's as indicated.

21. Remove any excess adhesive.
22. Install and tack weld the run-on/run-off tab's to the B-pillar outer panel.

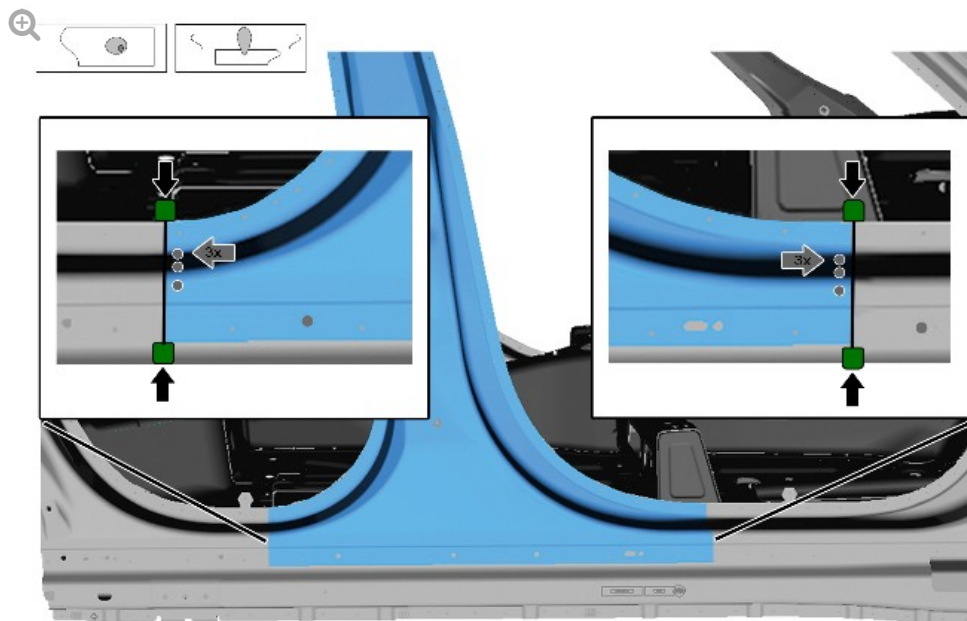
23.



E 144289

MIG plug weld the backing plate and MIG weld the B-pillar outer panel butt joint as indicated.

24.



E 144290

MIG plug weld the backing plates and MIG weld the B-pillar outer panel butt joint as indicated.

25. Cut off the run on/run off tabs.

26. Carry out a non destructive crack inspection on the MIG butt joints.

If correct proceed to next step, if not, rectify and recheck before proceeding.

27. Dress the welded joints.

28. Install the upper and lower rear door hinges and tighten the bolts to 35Nm.

29. Make sure that any open or exposed panel joints are suitably sealed.

30. Make sure corrosion protection is applied to all areas affected by repair.

31. The installation of associated panels and components is the reversal of removal procedure.